

# Quantitative Methods for Business and Economics

## Lecture 3 - Equation Systems and Graphing

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# Equation systems

- A set of equations that share the same variables is called an *equation system*.
- For example:

$$\begin{cases} x + y = 3 & (1) \\ 2x - y = 1 & (2) \end{cases}$$

- Because both (1) and (2) share  $x$  and  $y$ , they form an equation system.
- We usually want to solve the system, i.e., find the values of  $x$  and  $y$  that satisfy both equations.

# Solving equation systems

- There are several methods to solve equation systems.
  - Substitution
  - Elimination
  - Graphing

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- Substitution is typically the most “mechanical” method.
  - Express one variable in terms of the other and substitute in the other equation.
- Elimination is more algebraic.
  - Add or subtract the equations to eliminate one variable.
  - Might involve multiplying one or both equations by a constant.

# Solving the example system

- Let's solve the example system:

$$\begin{aligned}x + y &= 3 & (1) \\2x - y &= 1 & (2)\end{aligned}$$

- We can solve this system by substitution.
  - From (1), we have  $y = 3 - x$ .
  - Substitute this into (2):

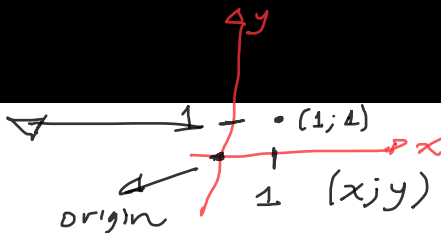
$$2x - (3 - x) = 1$$

- Solve for  $x$  and then substitute back to find  $y$ .

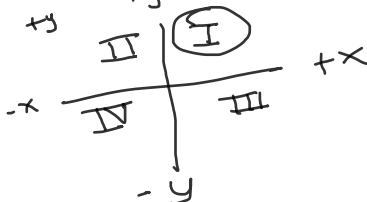
$$\frac{4}{3} + y = 3 \Rightarrow y = 3 - \frac{4}{3} = \frac{5}{3}$$

$$\begin{aligned}(1) \quad y &= 3 - x \\&\downarrow \text{into} \\(2) \quad 2x - (3 - x) &= 1 \\2x - 3 + x &= 1 \\3x - 3 &= 1 \\3x &= 4 \\x &= \frac{4}{3}\end{aligned}$$

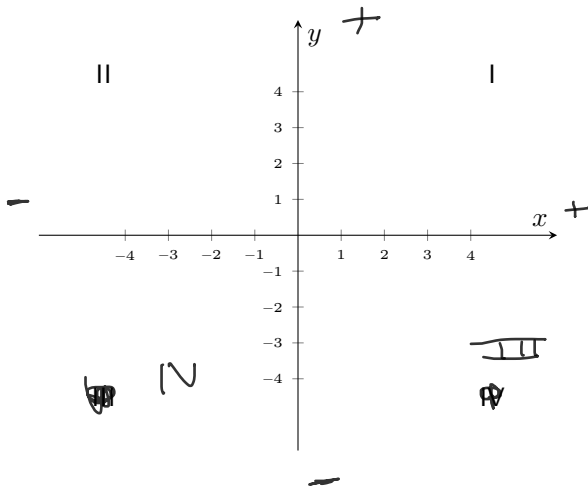
# The Cartesian plane



- The Cartesian plane is a two-dimensional space where we can plot points.
- It is formed by two perpendicular lines, the  $x$ -axis and the  $y$ -axis.
- The point where the axes intersect is called the *origin*.
- The axes divide the plane into four *quadrants*.

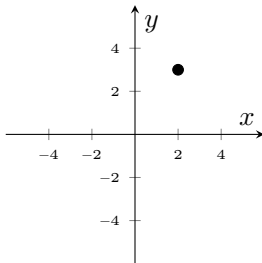


# The Cartesian plane



# Plotting points

- To plot a point, we use an ordered pair  $(x, y)$ .
  - $x$  is the distance from the  $y$ -axis.
  - $y$  is the distance from the  $x$ -axis.
- For example, the point  $(2, 3)$  is 2 units to the right and 3 units up from the origin. See below:



# Linear equations

Most

- 1 The equations we've seen so far are *linear* equations.
  - They represent straight lines in the Cartesian plane.
  - Linear equations can be written in the form  $y = mx + b$ .
    - $m$  is the slope of the line.
    - $b$  is the y-intercept.



$m \rightarrow \rightarrow$

steep

$m \approx 0$

$\hookrightarrow$  flat





# The Slope

- The ratio of the vertical change to the horizontal change.

- It tells us how steep the line is.  $\rightarrow m$

- The bigger the slope, the steeper the line.

- Given by  $m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{4 - 3}{3 - 2} = \frac{1}{1} = 1$

- Requires two points (call them  $P_1$  and  $P_2$ ) on the line, with coordinates  $(x_1, y_1)$  and  $(x_2, y_2)$ .

$$(2, 3) \text{ --- } (3, 4)$$

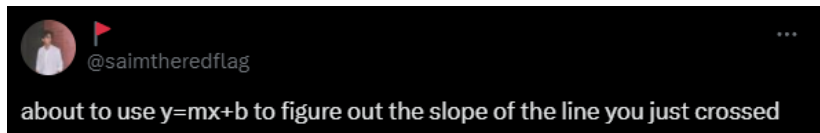
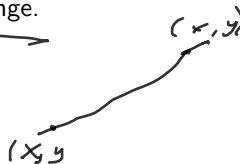
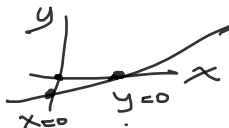


Figure 1: A meme

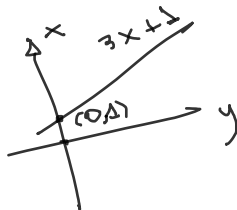
# Intercepts



- The *y-intercept* is the point where the line crosses the *y*-axis.
  - This happens when  $x = 0$ .
  - So, we set  $x = 0$  in the equation and solve for  $y$ .
  - In the equation  $y = mx + b$ , the *y*-intercept is  $(0, b)$ .
- The *x-intercept* is the point where the line crosses the *x*-axis.
  - This happens when  $y = 0$ .
  - So, we set  $y = 0$  in the equation and solve for  $x$ .

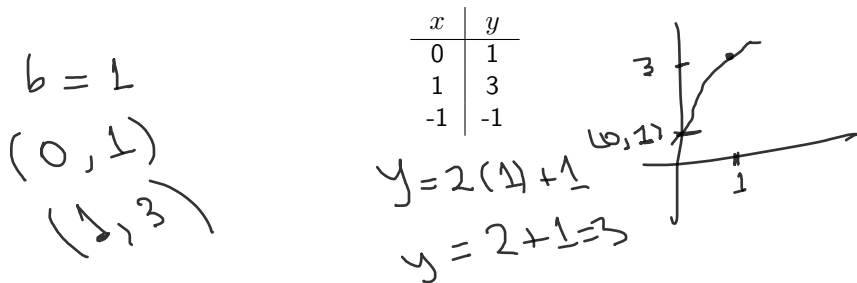
$$y = 3x + 1$$
$$mx + b$$

$$b=1 \quad m=3$$



# Graphing linear equations

- To graph a linear equation, we need to find two points on the line.
  - The easiest points are the intercepts.
  - We can also use the slope to find a second point.
- Example: graph the line  $y = 2x + 1$ .
  - It might be useful to draw a table of values.



# Graphing the line

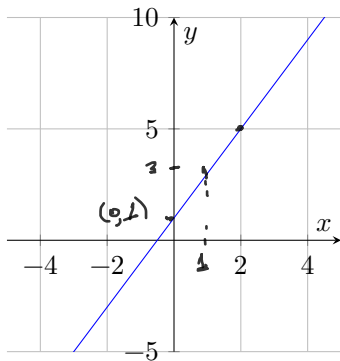


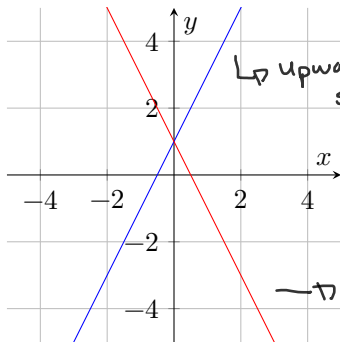
Figure 2: Plot of the equation  $y = 2x + 1$

# Upward-sloping and downward-sloping lines

- If  $m > 0$ , the line is “upward-sloping” or increasing.
  - As  $x$  increases,  $y$  also increases.
- If  $m < 0$ , the line is “downward-sloping” or decreasing.
  - As  $x$  increases,  $y$  decreases.

$$y = 1 - \frac{2}{3}x$$

$$m \leq 0 - \frac{2}{3}$$



↪ upward  
sloping

(+) x  
(+) y

↪ sales  
attribution

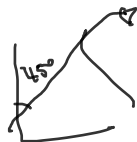
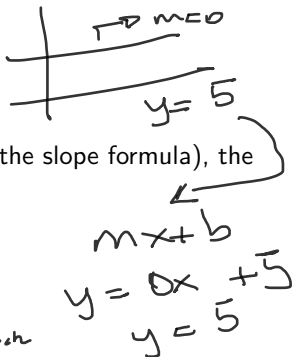
↪ down  
ward  
sloping

(+) x  
(-) y  
(+) P  
(-) Q

Figure 3: Upward-sloping and downward-sloping lines

# Properties of slopes

- If  $m = 0$ , the line is horizontal.
  - $y$  does not change as  $x$  changes.
- If the slope is undefined (division by zero in the slope formula), the line is vertical.
  - $x$  does not change as  $y$  changes.
- If  $m = 1$ , the line has a 45-degree angle.
- Lines with the same slope are parallel.  $\rightarrow$  touch
- Lines with slopes that multiply to  $-1$  are perpendicular.



■ This means that  $m_1 \cdot m_2 = -1$ , or that  $m_1 = -\frac{1}{m_2}$  (the negative reciprocal).

Handwritten calculations and diagrams:

- A coordinate system showing a vertical line at  $x=5$  and a horizontal line at  $y=1$ .
- A coordinate system showing two points:  $(1, 4)$  and  $(1, 3)$ , representing a vertical line.
- A calculation:  $m = \frac{4-3}{1-1} = \frac{1}{0}$ , with an arrow pointing to the word "undefined".

# How to find the equation of a line

$$y = mx + b$$

$m$  and  $P_1$   
 $P_1$  and  $P_2$

- 1 If you know the slope  $m$  and a point  $(x_1, y_1)$  on the line, you can use the point-slope form:

↪  $y - y_1 = m(x - x_1)$  ✓ "point slope"

- 2 If you know two points  $(x_1, y_1)$  and  $(x_2, y_2)$  on the line, you can use the slope formula to find  $m$  and then use  $y = mx + b$  to find  $b$ .

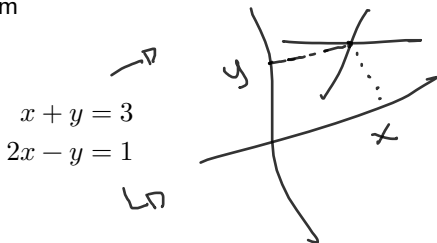
- 3 If you know the slope  $m$  and the  $y$ -intercept  $b$ , you can use  $y = mx + b$  directly (this is the slope-intercept form)

$$m = 3/4$$
$$b = -1/2$$

$$y = mx + b$$
$$y = \frac{3}{4}x - \frac{1}{2}$$

# Graphing equation systems

- To solve an equation system graphically, we graph both equations and find the point where they intersect.
- The point of intersection is the solution to the system.
- Example: graph the system





# Graphing the system

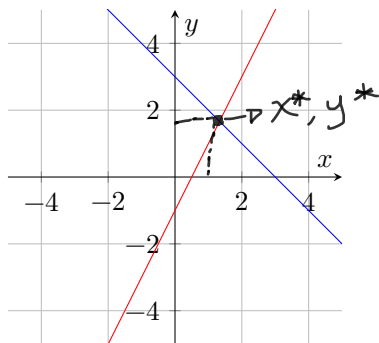


Figure 4: Graph of the system

# Systems with no solutions

$$m = \frac{5}{2}$$

$$y = \frac{5}{2}x + 3 \quad (1)$$

$$m = \frac{10}{4} = \frac{5}{2}$$

$$y = \frac{10}{4}x + 3 \quad (2)$$

- Sometimes, the lines are parallel and do not intersect.
  - This means that the system has no solution.
- How can we tell if two lines are parallel?
  - They have the same slope. ==
  - The coefficients of  $x$  and  $y$  in the equations are proportional.



$$y = \frac{2}{3}x + 5$$
$$y = \frac{2}{3}x + \frac{10}{5}$$

Handwritten annotations: An arrow points from the coefficient 2 in the first equation to the coefficient 2 in the second equation. Another arrow points from the constant term 5 in the first equation to the constant term 10/5 in the second equation, illustrating the simplification of the constant term.

# Systems with infinite solutions

- Sometimes, the lines coincide and intersect at every point.
  - This means that the system has infinite solutions.
- How can we tell if two lines coincide?
  - They have the same slope and the same  $y$ -intercept.
  - The coefficients of  $x$  and  $y$  in the equations are proportional, and the constants are equal.

# Graphic representations - systems with no solutions

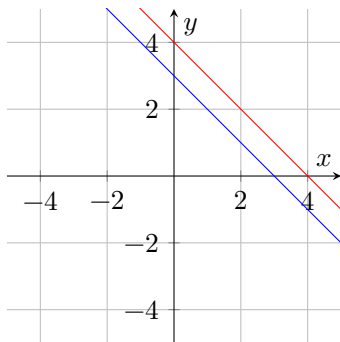


Figure 5: Graph of a system with no solutions

## Graphic representations - systems with infinite solutions

$$y = 2x + 1 \quad (1)$$

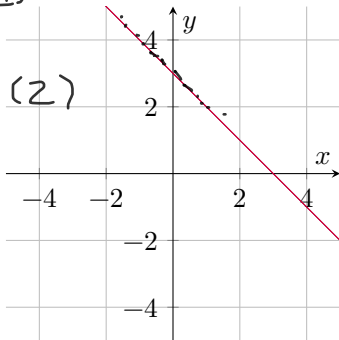
$$y = \frac{4}{2}x + \frac{13}{13} \quad (2)$$

$$m = 2$$

$$m = 2$$

$$b = 1$$

$$b = 1$$



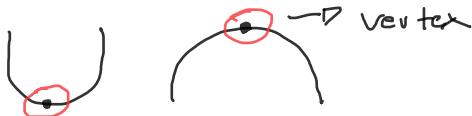
# What happens when I try to solve a system that can't be solved?

- If you try to solve a system that has no solution, you will get a contradiction.
  - For example, you might find  $0 = 1$ .
- If you try to solve a system that has infinite solutions, you will get an identity.
  - For example, you might find  $0 = 0$ .
- In both cases, you cannot reach something like we're used to, like  $x = 3$  and  $y = 2$ .

# More than two equations

- We can extend the concept of equation systems to more than two equations.
  - For example, a system of three equations in three variables.
- The same principles apply
  - Elimination
  - Substitution
  - Graphing
- For graphing, we need to consider more dimensions.
  - For a system of three equations in three variables, we need a three-dimensional space (3D), with  $x$ ,  $y$ , and  $z$  axes.

# Quadratic equations revisited



- Quadratic equations are equations of the form  $y = ax^2 + bx + c$ .
  - They represent parabolas in the Cartesian plane.

- The vertex of the parabola is given by  $x = -\frac{b}{2a}$ .
 

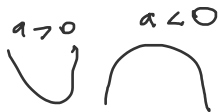
x-coordinate

- This is where the parabola reaches its maximum or minimum.

- The vertex is the point where the parabola changes direction.

- It has coordinates  $\left(\frac{-b}{2a}, f\left(\frac{-b}{2a}\right)\right) \rightarrow (x, y)$

- The parabola opens upwards if  $a > 0$  and downwards if  $a < 0$ .



$\rightarrow f(5) = a(5)^2 + b(5) + c$   
 $f(x) \Rightarrow$  plug in value of "x" in eqn.



# Graphing quadratic functions

(1) ■ To graph a quadratic function, the vertex is a good starting point.

(2) ■ If there are any roots or x-intercepts, they are also useful.

- The roots are the points where the parabola crosses the x-axis.

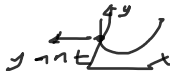
- They are given by the solutions to the equation  $ax^2 + bx + c = 0$ .

(3) ■ Intercepts with the y-axis might be useful too.

- The y-intercept is the point where the parabola crosses the y-axis.

- Solved by setting  $x = 0$  in the equation.

- They don't necessarily exist.



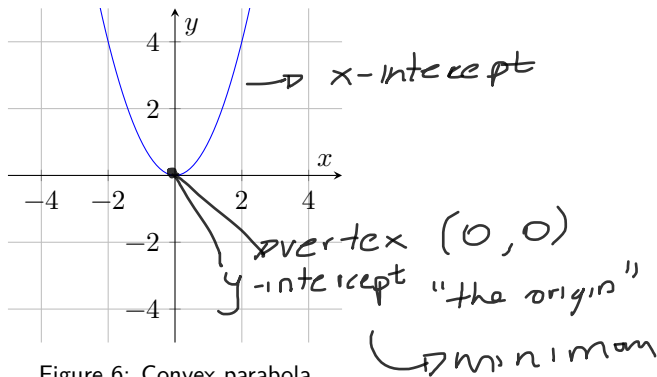
(4) ■ Ultimately, might need to use a table of values to plot the parabola.

| $x$ | $y$    |
|-----|--------|
| 1   | $f(1)$ |
| 2   | $f(2)$ |

  $\rightarrow f(x) = ax^2 + bx + c$

# Convex parabola

- A parabola that opens upwards is called convex.
- It has a minimum at the vertex.



# Concave parabola

- A parabola that opens downwards is called *concave*.
- It has a maximum at the vertex.

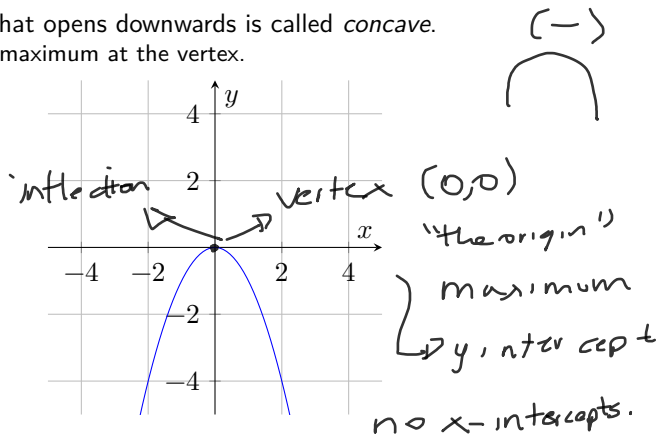


Figure 7: Concave parabola

# Cubic equations/functions

- Cubic equations are equations of the form  $y = ax^3 + bx^2 + cx + d$ .
  - They represent cubic functions in the Cartesian plane.
- The graph of a cubic function is a curve with an inflection point.
- An inflection point is a point where the curve changes concavity.
  - Convex to concave or vice versa.



↳ direction  
"up"  
"down"

# Graph of a cubic function

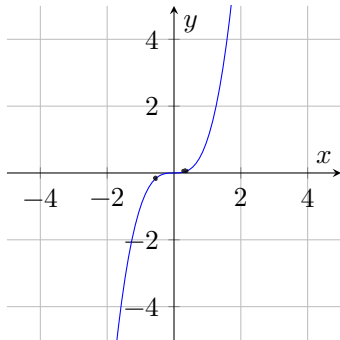


Figure 8: Graph of  $x^3$

# Application: comparing cost structures

- A firm needs machinery and can rent from two suppliers:
  - **Supplier A:** \$2,000 flat fee plus \$50 per hour of use
  - **Supplier B:** \$500 flat fee plus \$80 per hour of use
- Write each total cost as a linear equation in hours of use  $h$ :

$$C_A = 2,000 + 50h$$

$$C_B = 500 + 80h$$

- Which supplier is cheaper depends on how many hours the firm needs
- The two cost lines intersect at the **indifference point**: the usage level where both suppliers cost the same

## Application: comparing cost structures

- Set  $C_A = C_B$  and solve the system:

$$2,000 + 50h = 500 + 80h$$

$$1,500 = 30h$$

$$h^* = 50 \text{ hours}$$

- Below 50 hours, Supplier B is cheaper (lower flat fee)
- Above 50 hours, Supplier A is cheaper (lower hourly rate)
- Graphically, this is exactly the intersection of two lines — the same equation systems we solved earlier

## Example: comparing cost structures

- For you: a gym charges a \$30 flat monthly fee. A second gym charges a \$10 monthly fee plus \$2 per visit.
- How many visits per month make both gyms equally costly?
- Which gym should a member who visits 8 times per month choose?